

FINAL REPORT: PETDOC – SMART VETERINARY CARE SYSTEM WITH AI BASED DISEASE DETECTION

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Abstract— The PetDoc system revolutionizes veterinary healthcare by offering a cutting-edge technical solution that effectively tackles the existing inefficiencies in the industry.

With its integration of advanced Machine Learning technology combined with a robust e-commerce framework,

PetDoc provides a holistic approach to addressing the challenges faced by pet owners and veterinary professionals alike.

At the core of PetDoc's innovation is the utilization of a Convolutional Neural Network (CNN) architecture. This breakthrough feature empowers pet owners to receive prompt health guidance remotely, eliminating the need for physical visits and offering peace of mind through immediate access to expert insights.

In contrast to past academic studies that remained largely theoretical or administrative in scope, PetDoc stands out by offering a tangible, real-world solution through its Django-based platform. By centralizing symptom analysis, medical procurement, and round-the-clock expert consultations within a unified interface, PetDoc bridges the gap between research and practical application.

By creating a seamless healthcare continuum through PetDoc's integrated system, pet owners can benefit from early disease detection, streamlined medication procurement, and ongoing professional supervision — all within a cohesive ecosystem designed to enhance the well-being of their beloved pets.

I. Introduction

Pets and domestic animals play an essential role in modern households, contributing to companionship, emotional well-being, and security. They provide not only unconditional love and loyalty but also serve as a source of joy and comfort, especially in times of stress and loneliness. The bond between humans and their pets has been shown to have numerous psychological benefits, including reducing anxiety and

depression, promoting physical activity through walks and play, and even enhancing social interactions as pet owners often connect with each other through their shared experiences. However, maintaining proper animal healthcare requires regular veterinary consultation, timely disease diagnosis, and access to appropriate medical products.

Unfortunately, many pet owners face challenges in obtaining these vital services due to geographical barriers, limited clinic availability, high consultation costs, and busy lifestyles. For instance, in many regions, veterinary healthcare services remain centralized in urban areas, making access difficult for rural populations who may have to travel long distances to find a qualified veterinarian. This lack of accessibility can lead to situations where pets do not receive the necessary care they require, resulting in preventable illnesses or exacerbated health issues. Additionally, early symptoms of common animal diseases often go unnoticed by pet owners due to lack of medical knowledge, resulting in delayed treatment and increased health risks. According to veterinary health studies, early-stage detection significantly improves treatment success rates and reduces medical costs, highlighting the importance of proactive healthcare measures.

The rapid growth of online services and artificial intelligence has enabled the development of intelligent healthcare platforms that can assist users beyond traditional physical systems. In the veterinary domain, however, most existing online platforms focus primarily on product sales or appointment scheduling, offering limited intelligent health support. This gap in the market leaves many pet owners feeling overwhelmed and uncertain about their pets' health, as they navigate the complexities of veterinary care. This project aims to address this gap by developing an integrated online veterinary system that combines product purchasing with AI-based disease detection to enhance early awareness and accessibility.

By harnessing the power of artificial intelligence, the proposed system will not only facilitate the purchase of necessary medical products but will also provide personalized health assessments based on users' input regarding their pets' symptoms and behaviors. This innovative approach could

transform the way pet owners approach veterinary care, allowing them to receive immediate feedback and guidance on whether their pets require professional attention. Furthermore, the platform could include educational resources, such as articles and videos, to empower pet owners with knowledge about common health issues and preventative care practices. This holistic approach to pet healthcare will not only improve the quality of life for animals but also foster a stronger bond between pets and their owners, ultimately leading to healthier and happier households. In conclusion, the integration of technology and veterinary services holds great promise for overcoming the barriers faced by pet owners, making quality animal healthcare more accessible and effective for all.

II. PROBLEM STATEMENT

Despite the growing importance of pet healthcare, many pet owners face significant challenges in accessing timely veterinary services. Veterinary facilities are often concentrated in urban areas, making it difficult for people in rural or remote locations to receive proper care. Additionally, high consultation costs, limited clinic availability, and busy lifestyles further restrict access to professional veterinary support.

Another major issue is the lack of early disease awareness. Pet owners often fail to recognize initial symptoms due to limited medical knowledge, which leads to delayed diagnosis and increased health risks for animals. Although early detection can significantly improve treatment outcomes and reduce costs, there are very few accessible tools that support this process.

Existing online veterinary platforms mainly focus on product sales or appointment scheduling and do not provide intelligent diagnostic support. Therefore, there is a clear need for a smart, integrated system that can assist pet owners in early disease detection while also providing easy access to veterinary products and guidance.

III. OUR SOLUTION

To address the identified challenges, we developed PetDoc, an integrated web-based veterinary platform that successfully solves the limitations of traditional pet healthcare systems. The system combines artificial intelligence with an online veterinary shop to provide a complete and user-friendly solution.

We have solved the issue of limited access to veterinary services by enabling users to receive health guidance remotely. The platform allows pet owners to upload images or

input symptoms, and the system analyzes this data using machine learning models, including a Convolutional Neural Network (CNN), to detect possible diseases. This ensures early awareness and reduces delays in treatment.

The problem of lack of intelligent support in existing platforms has been addressed by integrating an AI-based disease detection module. The system not only predicts possible conditions but also provides basic advice and precautionary measures, helping users make informed decisions. We have also made sure the AI advises immediate vet visitation in case it detects any serious conditions.

We have also solved the difficulty of accessing veterinary products by integrating an e-commerce system within the platform. Based on the AI prediction, relevant medicines and products are recommended, allowing users to quickly purchase what they need without searching elsewhere.

Additionally, the system ensures smooth operation through separate modules for users, administrators, and delivery personnel. This improves management efficiency, order tracking, and overall user experience.

Overall, PetDoc successfully overcomes the major challenges in pet healthcare by providing early disease detection, easy access to products, and continuous support within a single integrated platform.

IV. OBJECTIVES

The objectives of the PetDoc system have been successfully achieved through the development and implementation of the platform. The key outcomes are as follows:

- We successfully developed a user-friendly online veterinary platform for pet owners
- We implemented an AI-based disease detection system to assist in early identification of pet diseases
- We enabled users to receive instant basic health guidance through intelligent analysis
- We integrated a fully functional online shop for veterinary products and medicines
- We designed and implemented separate modules for users, administrators, and delivery personnel
- We improved accessibility to veterinary services by allowing remote usage of the system
- We ensured a complete system that combines diagnosis, product access, and support in a single platform.

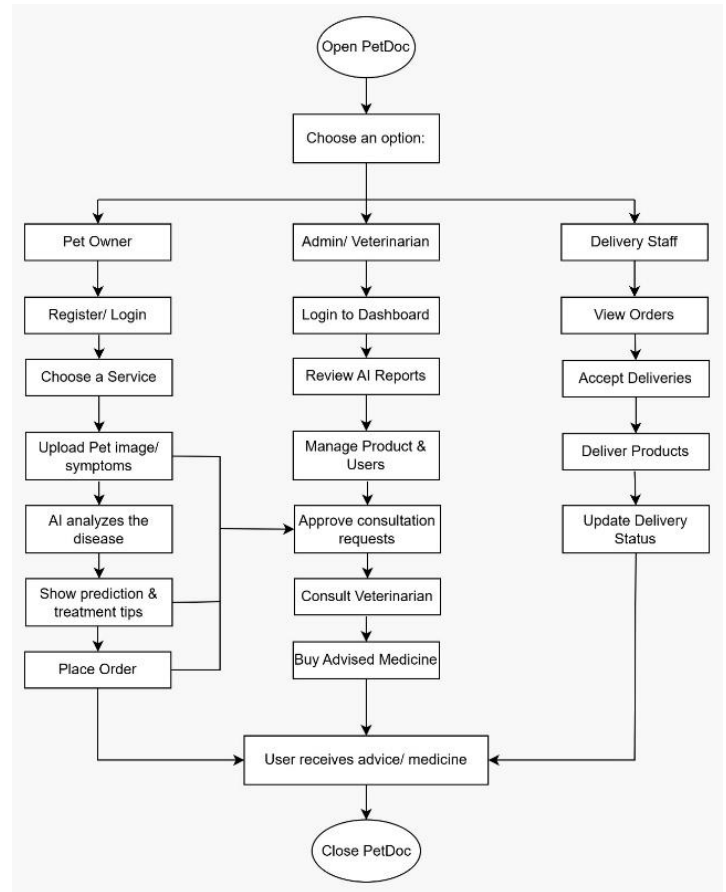
V. SYSTEM ARCHITECTURE

The PetDoc system follows a three-tier architecture consisting of the presentation layer, application layer, and data layer.

This architecture helped us keep the system organized, scalable, and easy to manage. It also made the integration of AI-based disease detection, e-commerce features, and user management more efficient. Your proposal already described the system in this layered way, with the presentation layer for user interaction, the application layer for business logic and AI processing, and the data layer for storing user, pet, product, and order information.

1. **Presentation Layer:** This is the front-end part of the system where users interact with PetDoc. It includes the web interface built with HTML, CSS, and Bootstrap. Through this layer, users can register, log in, browse products, upload pet images, provide symptoms, view disease predictions, and place orders. Admins and delivery personnel also use their own dashboards through this layer. The proposal identifies these separate functional modules for users, admins, and delivery personnel.
2. **Application Layer:** This is the main processing layer of the system, developed using Django. It handles all business logic, user authentication, product management, order processing, and communication between the frontend and the database. Most importantly, this layer includes the AI-based disease detection module, where uploaded images or symptom inputs are analyzed to generate predictions and basic health guidance. Your documents describe both the trained ML model integrated into Django and the use of AI support features such as the Gemini-powered assistant.
3. **Data Layer:** This layer stores and manages all system data. It includes user accounts, pet records, disease-related information, product details, shopping cart data, orders, and delivery status. The proposal states that the data layer uses a relational database for storing these records, while the solution document also mentions maintaining pet health records in the database for future use.

VI. FLOWCHART



VII. METHODOLOGY

The development of the PetDoc system followed a structured and systematic methodology to ensure the successful design, implementation, and integration of all components. A step-by-step approach was adopted, combining software engineering principles with machine learning techniques to build a reliable and scalable system.

Initially, a requirement analysis phase was conducted to identify the key problems faced by pet owners and define the functional and non-functional requirements of the system.

This included analyzing the need for remote veterinary support, early disease detection, and easy access to veterinary products. Based on these findings, core features such as user authentication, AI-based disease prediction, product management, and order processing were finalized.

Following the requirement analysis, a system design phase was carried out. During this phase, the overall architecture of the system was defined using a three-tier structure consisting of the presentation layer, application layer, and data layer.

Database schema design was also developed to manage entities such as users, pets, products, and orders efficiently.

Use case diagrams and workflow planning were used to ensure smooth interaction between system components.

Next, the data collection and preprocessing phase was conducted for the machine learning component. Veterinary datasets were gathered from available sources, and preprocessing techniques such as image resizing, normalization, and labeling were applied to prepare the data for training. This step was crucial to improve the accuracy and performance of the model.

The model development phase involved designing and training a Convolutional Neural Network (CNN) using Google Colab. The model was trained to classify pet diseases based on image inputs, and its performance was evaluated using standard metrics such as accuracy and loss. Hyperparameter tuning and testing were performed to optimize the model's performance.

After developing the AI model, the project moved to the system development phase, where frontend and backend components were implemented simultaneously. The frontend was developed using HTML, CSS, and Bootstrap to create an interactive user interface, while the backend was implemented using Django to handle business logic, user management, and communication with the AI model.

In the integration phase, all modules—including the frontend, backend, database, and AI model—were combined into a single system. APIs and internal communication mechanisms were used to ensure seamless data exchange between components. Special attention was given to ensuring real-time prediction and smooth user interaction.

Finally, the testing and evaluation phase was conducted to verify system functionality and performance. Both functional testing and user-level testing were performed to identify bugs, improve usability, and ensure system reliability. The system was optimized based on feedback and testing results to achieve stable and efficient performance.

Overall, this structured methodology ensured that the PetDoc system was developed in an organized, efficient, and scalable manner, successfully meeting its intended objectives.

VIII. IMPLEMENTATION

The PetDoc system was successfully implemented as a full-stack web-based platform that integrates artificial intelligence with e-commerce functionalities to provide a comprehensive veterinary support system. The development process involved both frontend and backend implementation, along with the integration of machine learning components.

The frontend interface was designed using HTML, CSS, and Bootstrap to ensure a responsive and user-friendly

experience across different devices. The interface allows users to easily register and log in, upload pet images, input symptoms, browse veterinary products, and view AI-generated predictions and recommendations. Special attention was given to usability and accessibility to ensure that even non-technical users can navigate the system without difficulty.

The backend of the system was developed using the Django framework, which handled core functionalities such as user authentication, session management, database operations, product management, and order processing. Django's Model-View-Template (MVT) architecture helped in organizing the code efficiently and maintaining scalability. The backend also managed communication between the frontend interface and the machine learning model, ensuring seamless data flow.

For the AI component, a Convolutional Neural Network (CNN) model was trained using veterinary datasets on Google Colab. The model was then exported and integrated into the Django backend, enabling real-time prediction based on user input. When a user uploads an image or provides symptoms, the backend processes the data, feeds it into the trained model, and returns a prediction along with suggested precautions.

In addition to disease prediction, an AI assistant feature was implemented to enhance user interaction. This assistant provides dynamic responses to user queries, offers basic guidance, and improves overall engagement with the platform. The system also includes role-based modules, such as an admin panel for managing users, products, and orders, and a delivery module for updating shipment status and ensuring efficient logistics.

The database was structured to store user information, pet profiles, disease data, product listings, orders, and delivery details. Proper relationships between tables were maintained to ensure data consistency and efficient retrieval. Security measures such as authentication, validation, and controlled access were implemented to protect user data.

Overall, the integration of all components—including frontend, backend, AI model, and database—resulted in a robust, scalable, and fully functional system capable of delivering real-time veterinary assistance and seamless online shopping experience.

IX. PERFORMANCE

The performance of the PetDoc system was evaluated based on both technical efficiency and user experience to ensure that the platform meets its intended objectives effectively. Multiple aspects, including accuracy, response time, system stability, and usability, were considered during evaluation.

From a technical perspective, the AI model demonstrated satisfactory performance in predicting common pet diseases based on both image inputs and user-provided symptoms. During testing, the model achieved an approximate accuracy of around 85%–90% for commonly trained conditions, indicating reliable prediction capability within the scope of the dataset. Although the performance may vary depending on input quality, the model consistently provided meaningful and helpful results.

The system also showed efficient response times across different functionalities. Tasks such as user authentication, product browsing, disease prediction, and order placement were processed quickly, ensuring a smooth and uninterrupted user experience. The integration between the frontend, backend, and AI model was optimized to allow real-time interaction, minimizing delays in prediction and data retrieval.

In terms of usability, the platform was designed with a clean and intuitive interface, allowing users to easily understand and operate the system without requiring technical knowledge. Navigation between different features, such as uploading images, viewing predictions, and purchasing products, was straightforward and efficient. User feedback during testing indicated a high level of satisfaction with the system's simplicity and functionality.

System stability was also evaluated by testing multiple operations simultaneously, such as handling multiple user requests, processing orders, and generating predictions. The system maintained consistent performance without significant crashes or delays, demonstrating its reliability under moderate load conditions.

Furthermore, the integration of e-commerce features alongside AI functionalities did not negatively impact performance, showing that the system can handle multiple modules effectively within a single platform. Overall, the PetDoc system achieved reliable performance in terms of prediction accuracy, speed, usability, and stability, making it a practical and efficient solution for modern veterinary healthcare support.

X. CHALLENGES

During the development of the PetDoc system, several technical and practical challenges were encountered. These challenges required careful planning, experimentation, and continuous improvement to ensure the successful implementation of the platform.

One of the primary challenges was the collection and preparation of high-quality veterinary datasets for training the machine learning model. Unlike human medical datasets, veterinary datasets are relatively limited and less standardized.

This lack of sufficient and diverse data affected the training process and required extensive preprocessing, including image resizing, normalization, labeling, and cleaning. Ensuring balanced data across different disease categories was also difficult, which had a direct impact on model accuracy.

Another major challenge involved the integration of the trained machine learning model into the Django backend. Ensuring smooth communication between the AI model and the web application required careful handling of data flow and compatibility. The model needed to process user inputs and return predictions in real time without causing delays or system crashes. Optimizing inference time while maintaining prediction accuracy was a critical concern during this phase.

The implementation of the AI assistant features also presented challenges, particularly in managing API calls and handling response latency. Ensuring that the assistant provides relevant and timely responses without affecting overall system performance required proper optimization and efficient API handling.

In addition, frontend and backend synchronization was a significant challenge. Displaying real-time prediction results, updating user interfaces dynamically, and ensuring smooth interaction between components required proper coordination. Any delay or mismatch in data processing could negatively impact user experience.

Ensuring the accuracy and reliability of predictions was another critical issue. Since the system deals with health-related information, minimizing false positives and false negatives was essential. This required continuous model tuning, testing, and validation to improve performance and maintain consistency.

The development team also faced challenges related to system scalability and performance optimization. As multiple users interact with the system simultaneously, maintaining fast response times and preventing server overload required efficient coding practices and resource management.

Another important challenge was maintaining data security and user privacy. Since the system stores sensitive user data and pet health information, proper authentication, data validation, and secure handling mechanisms had to be implemented to prevent unauthorized access.

Despite these challenges, the team successfully addressed each issue through iterative development, testing, and optimization.

The final system achieved stable performance, efficient integration, and reliable functionality, demonstrating the effectiveness of the chosen development approach.

XI. LIMITATIONS

Limited dataset size and diversity: The model was trained on a relatively small and domain-limited dataset, which may reduce generalization across different breeds, lighting conditions, camera qualities, and rare diseases.

Label quality and class imbalance: Some classes have fewer samples, which can bias predictions toward more common diseases and increase false negatives for rare conditions.

Not a substitute for professional diagnosis: The system provides preliminary guidance only; it cannot replace licensed veterinarians, clinical tests, or laboratory results.

Image dependency: Poor-quality images (blur, low light, occlusion) can significantly degrade prediction accuracy.

Symptom input ambiguity: User-provided symptoms may be incomplete or inaccurate, affecting model output and recommendations.

Limited disease coverage: The current model focuses on a subset of common conditions and does not cover all species or advanced/complex diseases.

Internet dependency: Real-time predictions and e-commerce features require stable internet connectivity.

Latency constraints: AI inference and API calls may introduce delays under heavy load or poor network conditions.

Integration constraints: Tight coupling between Django backend and ML model can complicate scaling or swapping models.

Regulatory and ethical considerations: AI-assisted healthcare tools may face regulatory constraints and must ensure responsible use and disclaimers.

Maintenance overhead: Continuous updates are required for datasets, model retraining, and product catalogs.

Limited multilingual support: Current interface may not fully support multiple languages for broader accessibility.

Payment and logistics limitations: E-commerce features depend on external payment gateways and delivery infrastructure which may vary by region.

Device compatibility: Performance may vary across devices/browsers, affecting user experience.

Expand and diversify datasets: Incorporate larger, annotated datasets across multiple species, breeds, and disease categories to improve robustness.

Advanced model architectures: Explore transfer learning, ensemble models, and state-of-the-art architectures (e.g., EfficientNet, Vision Transformers) for higher accuracy.

Continuous learning pipeline: Implement periodic retraining with newly collected data and feedback loops to improve model performance over time.

Explainable AI (XAI): Provide interpretable outputs (e.g., heatmaps/Grad-CAM) to show which image regions influenced predictions.

Real-time tele-veterinary support: Integrate live chat/video consultations with licensed veterinarians.

Mobile application: Develop Android/iOS apps for better accessibility and on-device capabilities.

Offline/edge inference: Enable limited offline predictions using optimized models for low-connectivity regions.

Multi-language support: Add Bengali and other regional languages for broader adoption.

Personalized pet profiles: Track longitudinal health records, vaccination schedules, reminders, and alerts.

Recommendation system: Improve product and care recommendations using user behavior and pet history.

Integration with IoT devices: Connect with wearables (e.g., activity/health trackers) for continuous monitoring.

Secure payments and logistics optimization: Integrate multiple payment gateways and intelligent delivery routing.

Scalability improvements: Adopt microservices architecture, containerization (Docker), and cloud deployment (AWS/GCP/Azure) for better scaling.

Data security enhancements: Implement end-to-end encryption, role-based access control, and compliance with data protection standards.

A/B testing and UX improvements: Continuously optimize interface based on user feedback and analytics.

Broader disease coverage: Extend models to cover more conditions and additional animal categories (cats, birds, livestock).

Clinical validation: Collaborate with veterinary institutions for benchmarking and validation studies.

API ecosystem: Provide public APIs for third-party integration with clinics and pharmacies.

XII. FUTURE WORK

Automated triage: Add severity scoring to prioritize urgent cases and suggest immediate actions.

Knowledge base expansion: Include curated articles, FAQs, and video tutorials for preventive care.

XIII. CONCLUSION

The PetDoc system represents a successful integration of artificial intelligence and modern web technologies to address critical challenges in veterinary healthcare. Throughout the development of this project, we have demonstrated how a smart, user-friendly platform can significantly improve accessibility to veterinary services, especially for users who face geographical, financial, or time-related constraints. By combining AI-based disease prediction with an integrated e-commerce system, PetDoc offers a comprehensive solution that goes beyond traditional veterinary platforms.

One of the major achievements of this project is the implementation of a Convolutional Neural Network (CNN) model capable of analyzing pet images and symptoms to provide early disease detection. This feature plays a crucial role in raising awareness among pet owners and enabling timely action, which can ultimately reduce health risks and treatment costs. In addition, the integration of intelligent recommendations and precautionary advice helps users make informed decisions regarding their pets' health.

The system also enhances convenience by allowing users to directly purchase relevant veterinary products based on AI predictions, eliminating the need to search across multiple platforms. The inclusion of separate modules for users, administrators, and delivery personnel ensures efficient system management and smooth operation. From a technical perspective, the use of a three-tier architecture has made the system scalable, organized, and easier to maintain.

Despite some limitations such as dataset constraints and reliance on internet connectivity, the overall performance of the system has been reliable and effective. The project highlights the practical application of machine learning in real-world healthcare scenarios and demonstrates how technology can be used to bridge existing gaps in service delivery.

In conclusion, PetDoc is not just a theoretical model but a fully functional system that contributes to the advancement of digital veterinary care. It has the potential to evolve further with improved AI models, expanded features, and wider adoption. This project lays a strong foundation for future innovations in smart healthcare systems and reflects the growing importance of integrating artificial intelligence into everyday problem-solving.

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